

Artificial Intel Applications-20513 ITAI-2372

RT 2026 Section 413 3 Credits 08/17/2026 to 12/13/2026 Modified 08/12/2026

Our Vision

Houston City College will deliver relevant, high-quality education and training, ensuring success for all students, our community, economy and beyond. [About HCC \(https://www.hccs.edu/about-hcc/\)](https://www.hccs.edu/about-hcc/).

Course Meetings

Course Modality

(H) Hybrid. In Person Friday 6:00 PM to 8:50 PM

Meeting Days

Every Friday

Meeting Times

6:00 PM to 8:50 PM

Meeting Location

HCC West Loop Campus RM 155

Welcome and Instructor Information

What's Exciting About This Course

This course explores how artificial intelligence is applied to real-world processes across industries such as healthcare, finance, manufacturing, retail, transportation, and government. Through demonstrations, case studies, and practical assignments, students will learn to identify appropriate AI use cases, evaluate their risks and benefits, and develop a simple AI application.

My Personal Welcome

Welcome to ITAI 2372: Artificial Intelligence Applications. I look forward to working with you as we examine how AI is changing industries and the workplace. Throughout the course, I encourage you to ask questions, participate in discussions, and apply course concepts to areas that interest you.

Preferred Method of Contact

yuexin.mao@hccs.edu

Course Overview

Course Description

This is an intermediate course on the implications of intelligent technologies, A.I., and their impact on business and society. Students will examine specific applications of A.I. in business and the case histories of those implementations. This course will cover applications that range from healthcare to Oil & Gas,

Autonomous Transportation, Manufacturing, Digital Twins, Regional Emergency Response and Supply Chain/Logistics. The goal of this course is to have students learn how to apply A.I. in almost any type of environment where people need to peacefully coexist with technology, even in space.

This course focuses on the practical application of artificial intelligence within real-world business, industrial, and public-sector processes. Students examine how AI systems are selected, evaluated, integrated, and governed across industries such as healthcare, manufacturing, logistics, finance, government, agriculture, and transportation. Emphasis is placed on understanding how work actually happens in organizations, identifying decision points within workflows, and determining where AI can responsibly improve efficiency, accuracy, and outcomes. Through hands-on labs, structured case analyses, process-mapping exercises, and applied projects, students develop workforce-ready skills aligned with AI analyst, AI implementation, and AI operations roles

Requisites

[MATH 1314](#), [COSC 1436](#), [ITAI 1370](#), [ITAI 1371](#) [ITAI 1372](#). and ITAI 1378

Department Website

<https://www.hccs.edu/programs/areas-of-study/science-technology-engineering-math/artificial-intelligence/>

Core Curriculum Objectives (CCOs)

ITAI 2372 satisfies Component Area Option in the HCCS core curriculum. The HCCS Core Curriculum Committee has specified that the course address the following core objectives:

- *Critical Thinking*: Students will demonstrate the ability to engage in inquiry and analysis, evaluation and synthesis of information, and creative thinking by completing the class project and assignments, along with answering questions on quizzes and exams.
- *Communication Skills*: Students will demonstrate effective development, interpretation and expression of ideas through written, oral, and visual communication by completing assignments and participating in online or in-class discussions.
- *Quantitative and Empirical Literacy*: Students will demonstrate the ability to draw conclusions based on the systematic analysis of topics using observation, experiment, and/or numerical skills by completing assignments, and answering questions on quizzes and exams.

Student Learning Outcomes and Objectives

Program Student Learning Outcomes (PSLOs)

Can be found at: [Artificial Intelligence \(https://www.hccs.edu/programs/areas-of-study/science-technology-engineering-math/artificial-intelligence/%20-%20HCC%20\(hccs.edu\)\)](https://www.hccs.edu/programs/areas-of-study/science-technology-engineering-math/artificial-intelligence/%20-%20HCC%20(hccs.edu))).

Course Specific Student Learning Outcomes (CSLOs)

Upon completion of ITAI 2372 students will be able to:

1. Explain how AI systems operate within real-world organizational workflows, distinguishing between AI, automation, and traditional software tools.
2. Map business or operational processes and identify AI-relevant decision points using standardized process-mapping techniques.
3. Evaluate AI solutions used in industry using structured criteria including data requirements, decision impact, operational risk, and business value.
4. Analyze ethical, legal, and societal implications as operational risks in AI systems and recommend mitigation strategies.
5. Propose applied AI implementations that align with business needs, constraints, and workforce requirements.
6. Construct and present a comprehensive applied AI case analysis or implementation proposal demonstrating industry-ready documentation standards

Learning Objectives

By the end of this course students will be able to:

- Analyze real-world business and operational processes to identify opportunities for AI integration.
- Evaluate AI applications used in industry based on effectiveness, feasibility, risk, and impact.
- Interpret AI case histories to distinguish successful implementations from failed deployments.
- Apply ethical, legal, and regulatory considerations as practical constraints in AI deployment.
- Communicate AI implementation decisions to technical and non-technical stakeholders.
- Design and propose applied AI solutions aligned with organizational goals and workflows.

Departmental Practices and Procedures

Department Specific Instructor and Student Responsibilities: Artificial Intelligence Department

As your Instructor, it is my responsibility to:

- Provide the grading scale and detailed grading formula explaining how student grades are to be derived
 - Facilitate an effective learning environment through learner-centered instructional techniques
 - Provide a description of assignments Inform students of policies such as attendance, withdrawal, tardiness, and making up assignments
 - Provide the course outline and class calendar
- Arrange to meet with individual students as needed

As a student, it is your responsibility to:

- Attend class in person and/or online
- Participate actively by reading and reviewing course material, interacting with classmates, and responding promptly in your communication with your instructor.
- Complete the required assignments and exams.
- Ask for help when there is a question or problem.
- Keep copies of all paperwork, including this syllabus, handouts, and all assignments.
- Be aware of and comply with academic honesty policies in the HCCS Student Handbook.

Program-Specific Student Success Information : Artificial Intelligence Department

Expect to spend at least twice as many hours per week outside of class as you do in class studying the course content. Additional time is required for the assignments. The assignments provided will help you practice the concepts discussed in class lectures and hone your hands-on skills.

Successful completion of this course requires a combination of the following:

- Reading the materials provided from the Internet
- Critically viewing the videos provided from the Internet
- Attending class in person and/or online
- Working with your teammates to complete assignments
- Participating in class activities
- There is no short cut for success in this course; it requires reading and studying the material and more importantly completing the assignments.

Instructional Materials and Resources

Instructional Materials

The [HCC Online Bookstore \(https://hccs.bncollege.com/shop/hccs-central/page/find-textbooks\)](https://hccs.bncollege.com/shop/hccs-central/page/find-textbooks) provides searchable information on textbooks for all courses. Check with your instructor before purchasing textbooks because the book might be included in your course fees.

Add Instructional Materials Here

✓ Course Requirements

Assignments, Exams, and Activities

Type	Weight	Topic	Notes
Assignment	300 Points (30%)	Topic	Describe the assignment here.
Exams/Quizzes	150 Points (15%)	Topic	Describe the assignment here.
In-Class Activities	250 Points (25%)	Topic	Describe the assignment here.
Final Project	300 Points (30%)	Topic	Describe the assignment here.
Extra Credit	20 Points	Topic	Describe the assignment here.

Grading Formula

Grade	Range	Notes
A	900+ Points (90%+)	Excellent
B	800-899 Points (80-89.9%)	Good
C	700-799 Points (70-79.9%)	Average
D	600-699 Points (60-69.9%)	Passing
F	< 600 Points (< 60%)	Failing

* Instructor's Practices and Procedures

Incomplete Policy

In order to receive a grade of Incomplete ("I"), a student must have completed at least 85% of the work in the course. In all cases, the instructor reserves the right to decline a student’s request to receive a grade of Incomplete.

Missed Assignments/Make-Up Policy

I understand that unexpected circumstances can make it difficult to meet deadlines. For this reason, the course has a flexible submission policy:

- You may submit any assignment at any time until the last day of the course.
- However, on-time submissions are strongly encouraged because they receive the full score (100%)

Late Submission Penalty

To keep things fair and motivate timely work:

- For every 5-day delay after the original due date, 10 points will be deducted from your score for that assignment.

- Example:
 - Submitted on time → **100 points**
 - 6 days late → **90 points**
 - 11 days late → **80 points**, and so on.

Why This Policy?

This approach gives you flexibility while still rewarding punctuality. It ensures you can complete the course successfully even if you fall behind, but it also emphasizes the importance of staying on schedule.

Academic Integrity

Academic Integrity and Responsible Use of Generative AI

Academic integrity is essential to meaningful learning. In this course, you are encouraged to use generative AI tools to support brainstorming, research, and learning. However, all submitted work must demonstrate your own understanding, analysis, and judgment.

Copying and submitting AI-generated content without reviewing it, understanding it, or adding your own original thinking does not meet course expectations and may be considered a violation of academic integrity.

How to Use Generative AI Responsibly

- Use AI tools to explore ideas, clarify concepts, support research, or generate examples.
- Review and verify AI-generated information using reliable sources.
- Write in your own words and include your own analysis and conclusions.
- Cite AI-generated material when appropriate or required.
- Do not submit content that you cannot explain or defend as your own work.

Your work should reflect your individual learning and critical thinking. The purpose of this course is not simply to complete assignments, but to develop your ability to understand, evaluate, and apply AI responsibly.

Here's the link to the HCC information about academic integrity (Scholastic Dishonesty and Violation of Academic Scholastic Dishonesty and Grievance): [HCC Student Procedures](https://www.hccs.edu/studentprocedures) (<https://www.hccs.edu/studentprocedures>).

Attendance Procedures

WW – Online Classes

For fully online (WW) courses, attendance is measured by timely completion of assignments and tests. Students are expected to submit all work by the scheduled due dates to remain in good standing. Consistent engagement through assignments is essential for success in this course.

H – Hybrid Classes

For hybrid (H) courses, students must:

- Attend all scheduled in-person sessions
- Complete assignments and tests on time

Active participation during face-to-face meetings is a key component of the learning experience. Please plan accordingly to avoid disruptions to your progress.

Important:

- The first 3 absences will automatically result in the student being dropped from the course.
- Incomplete coursework or frequent absences can trigger Early Alerts, which may lead to subsequent withdrawal from the class.

Student Conduct

Be Prepared, Respectful, and Engaged ! Happy Learning !

Instructor’s Course-Specific Information

Assignments are due every Tuesday midnight, and are graded before next Tuesday.

Faculty Statement about Student Success

- **Stay Organized and Plan Ahead**
 - Review the syllabus and assignment deadlines early.
 - Use a calendar or planner to track due dates and in-person sessions (for Hybrid classes).
- **Engage Actively**
 - Participate in discussions, whether online or in class.
 - Ask questions and share your thoughts—your perspective matters!
- **Submit Work on Time**
 - On-time submissions earn full credit and help you avoid penalties.
 - Late work is accepted until the end of the course, but delays can significantly reduce your score.
- **Use Generative AI Responsibly**
 - AI tools can help you brainstorm and learn, but **do not copy content directly**.
 - Always write in your own words and cite sources when needed.
- **Communicate Early and Often**
 - If you face challenges, reach out to me as soon as possible.
 - Early communication can prevent issues from escalating.
- **Attend Regularly (Hybrid Classes)**
 - Missing more than 3 in-person sessions will result in being dropped from the course.
 - Frequent absences or incomplete work may trigger early alerts.
- **Practice Academic Integrity**
 - Your original thinking is essential. Avoid plagiarism and uphold honesty in all submissions.
- **Leverage Resources**
 - Use college tutoring services, library resources, and online tools to strengthen your understanding.

Faculty-Specific Information Regarding Canvas

This course section will use [HCC Eagle Online - Canvas \(https://eagleonline.hccs.edu\)](https://eagleonline.hccs.edu) to supplement in-class assignments, exams, and activities.

HCCS Open Lab locations may be used to access the Internet and Canvas. For best performance, Canvas should be used on the current or first previous major release of Chrome, Firefox, Edge, or Safari. Because it's built using web standards, Canvas runs on Windows, Mac, Linux, iOS, Android, or any other device with a modern web browser.

Canvas only requires an operating system that can run the latest compatible web browsers. Your computer operating system should be kept up to date with the latest recommended security updates and upgrades.

Statement Regarding Classroom Environment

HCC does not discriminate on the basis of race, color, religion, sex, gender identity and expression, national origin, age, disability, sexual orientation, or veteran status. I fully support that commitment and, as such, will work to maintain a positive learning environment based upon open communication, mutual respect, and non-discrimination. In this course, we share in the creation and maintenance of a positive and safe learning environment. Part of this process includes acknowledging and embracing the differences among us in order to establish and reinforce that each one of us matters. I appreciate your suggestions about how to best maintain this environment of respect. If you experience any type of discrimination, please contact me and/or the Office of Equal Opportunity and Title IX at 713.718.8271.



HCC Policies and Information

HCC Grading System

HCC uses the following standard grading system:

Grade	Grade Interpretation	Grade Points
A	Excellent (90-100)	4
B	Good (80-89)	3
C	Fair (70-79)	2
D	Passing (60-69), except in developmental courses.	1
F	Failing (59 and below)	0
FX	Failing due to non-attendance	0
W	Withdrawn	0
I	Incomplete	0
AUD	Audit	0
IP	In Progress. Given only in certain developmental courses. A student must re-enroll to receive credit.	0
COM	Completed. Given in non-credit and continuing education courses.	0

All grade points listed in the table above are per semester hour.

Note that most Health Sciences programs do not use the grading scale listed above. Program-specific grading scales can be found in the specific program handbook or course syllabus. For questions related to grading scales, students may contact the Dean of the College of Health Sciences or their Program Director.

Link to Policies in Catalog and Student Handbook

Here’s the link to the HCC Catalog and Student Handbook: [HCC Catalog \(https://catalog.hccs.edu/\)](https://catalog.hccs.edu/).

In it you will find information about the following:

- Academic Information
- Academic Support
- Attendance, Repeating Courses, and Withdrawal
- Career Planning and Job Search
- Childcare
- Ability Support Services
- Electronic Devices
- Equal Educational Opportunity
- Financial Aid TV (FATV)
- General Student Complaints
- Grade of FX
- Incomplete Grades
- International Student Services
- Health Awareness
- Libraries/Bookstore
- Police Services & Campus Safety

- Student Life at HCC
- Student Rights and Responsibilities
- Student Services
- Testing
- Transfer Planning
- Veteran Services

Link to HCC Academic Integrity Statement

[HCC Student Conduct \(https://catalog.hccs.edu/content.php?catoid=24&navoid=1706\)](https://catalog.hccs.edu/content.php?catoid=24&navoid=1706) (scroll down to subsections)

Campus Carry Link

Here's the link to the HCC information about Campus Carry:

[HCC Campus Carry \(https://www.hccs.edu/campuscarry\)](https://www.hccs.edu/campuscarry)

HCC Email Policy

When communicating via email, HCC requires students to communicate only through the HCC email system to protect your privacy. If you have not activated your HCC student email account, you can go [to HCC Eagle ID \(https://www.hccs.edu/email\)](https://www.hccs.edu/email) and activate it now. You may also use Canvas Inbox to communicate.

Office of Equal Opportunity and Title IX

Use the following link to access the HCC Office of Equal Opportunity and Title IX:

[HCC OEO Title IX \(https://www.hccs.edu/oeotix\)](https://www.hccs.edu/oeotix)

Ability Services

HCC strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on your disability (including long and short term conditions, mental health, chronic or temporary medical conditions), please meet with a campus Abilities Counselor as soon as possible in order to establish reasonable accommodations. Reasonable accommodations are established through an interactive process between you, your instructor(s) and Ability Services. It is the policy and practice of HCC to create inclusive and accessible learning environments consistent with federal and state law. For more information, please go to the following website: [HCC Accommodations \(https://www.hccs.edu/accommodations\)](https://www.hccs.edu/accommodations)

Title IX

Houston City College is committed to cultivating an environment free from inappropriate conduct of a sexual or gender-based nature including sex discrimination, sexual assault, sexual harassment, and sexual violence. Sex discrimination includes all forms of sexual and gender-based misconduct and violates an individual's fundamental rights and personal dignity. Title IX prohibits discrimination on the basis of sex-including pregnancy and parental status in educational programs and activities. If you require an accommodation due to pregnancy please contact an Abilities Services Counselor. The Director of EEO/Compliance is designated as the Title IX Coordinator and Section 504 Coordinator. All inquiries concerning HCC policies, compliance with applicable laws, statutes, and regulations (such as Title VI, Title IX, and Section 504), and complaints may be directed to:

Sandra Jacobson, J.D., M.Ed., SHRM-SCP Interim Director of EEO and Compliance/Title IX and ADA/Section 504 Coordinator Office of Equal Opportunity and Title IX

3100 Main, 7th Floor

Houston, TX 77002

713.718.8271

hcc.oeotix@hccs.edu (<mailto:hcc.oeotix@hccs.edu>),

[HCC OEO Title IX Website](https://www.hccs.edu/oeotix) (<https://www.hccs.edu/oeotix>).

Mandatory Reporters

Under Texas Education Code 51.252 (formerly known as Senate Bill 212), HCC Instructors are mandatory reporters of sexual harassment, dating violence (domestic violence), sexual assault, and stalking. All instructors are required by law to report to the College's Title IX coordinator or Deputy Title IX coordinator all reports disclosed to them relating to sexual harassment, dating violence (domestic violence), sexual assault, and stalking alleged to have been committed by or against a person who was a student enrolled at or an employee of the institution at the time of the incident. Instructors are required by law to include all the information they know about the incident, including the name of the student(s), in the report to the College's Title IX coordinator or deputy Title IX coordinator.

Office of the Dean of Students

Contact the office of the Dean of Students to seek assistance in determining the correct complaint procedure to follow or to identify the appropriate academic dean or supervisor for informal resolution of complaints.

[HCC Dean of Students](https://www.hccs.edu/policies-procedures-financial-transparency--disclosures/student-rights-responsibilities/speak-with-the-dean-of-students/) (<https://www.hccs.edu/policies-procedures-financial-transparency--disclosures/student-rights-responsibilities/speak-with-the-dean-of-students/>).

Student Success

Expect to spend at least twice as many hours per week outside of class as you do in class studying the course content. Additional time will be required for written assignments. The assignments provided will help you use your study hours wisely. Successful completion of this course requires a combination of the following:

- Reading the textbook
- Attending class in person and/or online
- Completing assignments
- Participating in class activities

There is no short cut for success in this course; it requires reading (and probably re-reading) and studying the material using the course objectives as a guide.

Canvas Learning Management System

Canvas is HCC's Learning Management System (LMS), and can be accessed at the following URL:

[HCC Eagle Online - Canvas](https://eagleonline.hccs.edu) (<https://eagleonline.hccs.edu>).

HCCS Open Lab locations may be used to access the Internet and Canvas. For best performance, Canvas should be used on the current or first previous major release of Chrome, Firefox, Edge, or Safari. Because it's built using web standards, Canvas runs on Windows, Mac, Linux, iOS, Android, or any other device with a modern web browser.

Canvas only requires an operating system that can run the latest compatible web browsers. Your computer operating system should be kept up to date with the latest recommended security updates and upgrades.

HCC Online Information and Policies

Here is the link to information about HCC Online classes, which includes access to the required Online Information Class Preview for all fully online classes: [HCC Online](https://www.hccs.edu/online) (<https://www.hccs.edu/online>).

Scoring Rubrics, Sample Assignments, etc.

Look in Canvas for the scoring rubrics for assignments, samples of class assignments, and other information to assist you in the course: [HCC Eagle Online - Canvas](https://eagleonline.hccs.edu) (<https://eagleonline.hccs.edu>).

Instructor and Student Responsibilities

As your Instructor, it is my responsibility to:

- Provide the grading scale and detailed grading formula explaining how student grades are to be derived
- Facilitate an effective learning environment through learner-centered instructional techniques
- Provide a description of any special projects or assignments
- Inform students of policies such as attendance, withdrawal, tardiness, and making up assignments
- Provide the course outline and class calendar that will include a description of any special projects or assignments
- Arrange to meet with individual students during office hours, and before and after class as required

As a student, it is your responsibility to:

- Attend class in person and/or online
- Participate actively by reviewing course material, interacting with classmates, and responding promptly in your communication with me
- Read and comprehend the textbook
- Complete the required assignments and exams
- Ask for help when there is a question or problem
- Keep copies of all paperwork, including this syllabus, handouts, and all assignments
- Be aware of and comply with academic honesty policies in the [HCCS Student Handbook \(https://www.hccs.edu/policies-procedures-financial-transparency--disclosures/student-rights-responsibilities/student-handbook/\)](https://www.hccs.edu/policies-procedures-financial-transparency--disclosures/student-rights-responsibilities/student-handbook/).

Sensitive or Mature Course Content

In this college-level course, we may occasionally discuss sensitive or mature content. All members of the classroom environment, from your instructor to your fellow students, are expected to handle potentially controversial subjects with respect and consideration for one another's varied experiences and values.

EGLS3

The EGLS³ ([Evaluation for Greater Learning Student Survey System \(https://hccs.campuslabs.com/courseeval/\)](https://hccs.campuslabs.com/courseeval/)) will be available for most courses near the end of the term but before finals. Results are confidential and will be available to faculty and division chairs after the end of the term, after grades are due. Confidential means that responses to the survey will not be associated with your identity when provided to the faculty.

[Link to HCC EGLS3 \(https://www.hccs.edu/egls3\)](https://www.hccs.edu/egls3)

Housing and Food Assistance for Students

If you are experiencing any hardship related to food, shelter, mental health, or other basic needs areas, please visit the Basic Needs page for resources ([HCC Cares \(https://www.hccs.edu/cares\)](https://www.hccs.edu/cares)). You have the option to take the Basic Needs Questionnaire and ask to be contacted by a counselor for additional assistance or support ([HCC Basic Needs \(https://www.hccs.edu/basicneeds\)](https://www.hccs.edu/basicneeds)). Furthermore, please notify the professor if you are comfortable doing so.

Student Resources

Tutoring

HCC provides free and convenient academic support, in a large variety of subjects, to HCC students in both an online environment and in-person on campus. Tutoring is provided by HCC personnel in order to ensure that it is appropriate. Visit the HCC Tutoring Services website for more information at [HCC Tutoring \(https://www.hccs.edu/tutoring\)](https://www.hccs.edu/tutoring).

Libraries

The HCC Library System consists of 9 libraries and 6 Electronic Resource Centers (ERCs) that are inviting places to study and collaborate on projects. Librarians are available both at the libraries and online to show you how to locate and use the resources you need. The libraries maintain a large selection of electronic resources as well as collections of books, magazines, newspapers, and audiovisual materials. The portal to all libraries' resources and services is the HCCS library web page at [HCC Library \(https://library.hccs.edu\)](https://library.hccs.edu).

Supplementary Instruction

Supplemental Instruction is an academic enrichment and support program that uses peer-assisted study sessions to improve student retention and success in historically difficult courses. Peer Support is provided by students who have already succeeded in completion of the specified course, and who earned a grade of A or B. Find details at [HCC Supplemental Instruction \(https://www.hccs.edu/supplemental-instruction\)](https://www.hccs.edu/supplemental-instruction).

Basic Needs Resources:

[HCC Cares Basic Needs Resources \(https://www.hccs.edu/support-services/counseling/hcc-cares/basic-needs-resources/\)](https://www.hccs.edu/support-services/counseling/hcc-cares/basic-needs-resources/).

Student Basic Needs Application:

[HCC Basic Needs \(https://www.hccs.edu/basicneeds\)](https://www.hccs.edu/basicneeds).

Instructional Modalities

HCC offers many options for learning modalities and information may be obtained at [HCC College Your Way \(https://www.hccs.edu/programs--courses/flexible-learning-online-hybrid-weekend/\)](https://www.hccs.edu/programs--courses/flexible-learning-online-hybrid-weekend/).

All students, regardless of their selected learning options, will still have access to the support available to help with their success, including tutoring, student life, basic needs support, career and employment services, counseling and ability services, and supplemental instruction. HCC also provides financial aid options.

In-Person (P)

In-Person gives students the opportunity to participate in class in person.

Online on a Schedule (WS)

Online on a Schedule allows students to take classes online, but at scheduled dates and times. Instead of visiting the campus for class, students log into the class online at the specified time they selected during registration. This gives students a safe and flexible learning option that allows for more interaction with professors and classmates without coming to campus. Access to a computer and WIFI are necessary to complete this class successfully.

Online Anytime (WW)

Online Anytime gives students the flexibility to complete coursework throughout the semester at times that works best for them. When a student enrolls in Online Anytime at the beginning of a semester, the course instructor will share pre-determined deadlines and requirements, along with the syllabus and full details of all assignments that must be completed to successfully finish the course. This option-most like a traditional online course -gives students the flexibility of completing coursework without visiting the campus. Access to a computer and WIFI is necessary to complete this class successfully.

Hybrid (H)

Hybrid provides students the opportunity to attend a lecture class meeting half-time in person and half-time remotely.

Hybrid Lab (HL)

Hybrid Lab-Based courses will be conducted in person and remotely, allowing HCC students access to essential hands-on, skills-based learning experiences needed for success.

Copyright Statement

In order to uphold the integrity of the academic environment and protect and foster a cohesive learning environment for all, HCC prohibits the unauthorized use of course materials. Materials shared in this course are based on my professional knowledge and experience as an instructor and are presented in an educational context for the students in the course. Authorized use of course materials is limited to personal study or educational uses. Material should not be shared, distributed, or sold outside the course without permission. Students are also explicitly forbidden in all circumstances from plagiarizing or appropriating course materials. This includes but is not limited to publicly posting quizzes, essays, or other materials. This prohibition extends not only during this course, but after. Sharing of the materials in any context will be a violation of the HCC Student Code of Conduct and may subject the student to discipline, as well as any applicable civil or criminal liability. Consequences for unauthorized sharing, plagiarizing, or other methods of academic dishonesty may range from a 0 on the specified assignment and/or up to expulsion from Houston City College. Questions about this policy may be directed to me, your instructor, or to the Manager of Student Conduct and Academic Integrity.

Unauthorized Disclosure

"Unauthorized disclosure" occurs when any student provides instructional materials and/or assessments to other students in violation of a clear prohibition by the instructor. Examples include: posting assessment items to online sites such as Chegg or CourseHero; asking exam questions in forums like Reddit or Yahoo Answers; discussions of confidential question using Wechat or GroupMe, etc.

Our Vision

Houston City College will deliver relevant, high-quality education and training, ensuring success for all students, our community, economy and beyond.

 Course Calendar

Syllabus Modifications

The instructor reserves the right to modify the syllabus at any time during the semester and will promptly notify students in writing, typically by e-mail, of any such changes.

 Additional Information

Course Detailed Outline

Key topics covered in this course include:

Module 0: Course Introduction & Workforce Orientation

SLOs Addressed: *SLO 1*

- Course expectations, academic integrity, and applied learning approach
- Overview of AI roles in the workforce: AI Analyst, AI Technician, AI Implementation Support
- Introduction to process thinking for AI integration
- Course tools and portfolio requirements (GitHub setup)

Hands-On Activity: Map a familiar process (online shopping, ride-sharing, food delivery) and identify where decisions occur. Students will use a simple flowchart template to document inputs, processes, decision points, and outcomes.

Module 1: How Work Happens – Business Processes for AI

SLOs Addressed: *SLO 1, SLO 2*

- What a business process is (practical, non-theoretical approach)
- The process flow: Inputs → Process → Decision → Action → Outcome
- Where AI fits in workflows (and where it does not)
- Process documentation standards and notation basics

Lab: Process-mapping exercise for a healthcare, retail, or logistics workflow. Students will select an industry scenario and create a detailed process map identifying at least 5 decision points where AI could potentially add value.

Module 2: Introduction to AI Applications Across Industries

SLOs Addressed: *SLO 1, SLO 3*

- Overview of AI capabilities: what AI can and cannot do
- Distinguishing AI vs. automation vs. traditional software tools
- Common AI application patterns: classification, prediction, optimization, generation
- The AI Decision Framework: Data → Model → Decision → Action

Hands-On Activity: Match real business problems to appropriate AI approaches. Students will be given 10 business scenarios and must identify: (1) whether AI is appropriate, (2) what type of AI approach fits, and (3) what data would be required.

Module 3: AI in Healthcare

SLOs Addressed: *SLO 2, SLO 3, SLO 4*

- Healthcare workflows: patient intake, diagnostics, operations, and monitoring
- AI applications: medical imaging analysis, predictive analytics, clinical decision support
- Regulatory considerations: HIPAA, FDA approval processes
- Human-in-the-loop requirements in medical AI

Lab: Workflow analysis of an AI healthcare use case (decision support vs. automation). Students will analyze a real healthcare AI implementation, document the workflow, and evaluate whether the AI system appropriately balances automation with human oversight.

Module 4: AI in Finance & Risk Management

SLOs Addressed: *SLO 2, SLO 3, SLO 4*

- Financial industry workflows: fraud detection, credit decisions, customer service
- Model risk and regulatory constraints (Fair Lending, model governance)
- Explainability requirements in financial AI
- Case studies: successful and failed AI implementations in banking

Lab: Case teardown: Why an AI finance system succeeded or failed. Students will analyze a documented case using a structured evaluation framework covering data quality, model design, integration approach, and governance.

Module 5: AI in Manufacturing & Industrial Operations

SLOs Addressed: *SLO 2, SLO 3, SLO 5*

- Manufacturing workflows: predictive maintenance, quality control, process automation

- Industrial IoT and sensor data integration
- Human-in-the-loop systems and safety-critical AI
- ROI analysis for manufacturing AI implementations

Hands-On Activity: Process improvement proposal using AI for a manufacturing workflow. Students will select a manufacturing scenario and develop a one-page proposal identifying the problem, proposed AI solution, required data, expected benefits, and implementation risks.

Module 6: AI in Retail, Supply Chain & Logistics

SLOs Addressed: *SLO 2, SLO 3, SLO 5*

- Retail and supply chain workflows: demand forecasting, inventory management, personalization
- AI impact on operations and customer experience
- Integration challenges: legacy systems, data silos, real-time requirements
- Case studies: Amazon, Walmart, and emerging retail AI applications

Lab: AI decision-mapping for a supply chain workflow. Students will map a complete supply chain process and identify 3-5 decision points where AI is currently used or could be applied, documenting the type of AI, data requirements, and expected impact.

Module 7: AI in Transportation & Smart Infrastructure

SLOs Addressed: *SLO 2, SLO 3, SLO 4*

- Transportation workflows: traffic management, logistics routing, fleet optimization
- Autonomous systems: current state, limitations, and future trajectory
- Safety, accountability, and public impact considerations
- Smart city applications and infrastructure AI

Lab: Evaluate AI deployment risks in transportation systems. Students will analyze a transportation AI implementation and complete a risk assessment covering safety, reliability, accountability, and public acceptance factors.

Module 8: AI in Media, Content & Creative Industries

SLOs Addressed: *SLO 3, SLO 4*

- Content creation workflows: generation, curation, personalization, moderation
- Generative AI applications: text, image, video, audio
- Human oversight requirements and quality control
- Intellectual property, attribution, and misuse risks

Hands-On Activity: Analyze a generative AI workflow and identify control points. Students will map a content creation workflow that uses generative AI and identify: (1) human review checkpoints, (2) quality control measures, (3) potential failure modes, and (4) mitigation strategies.

Module 9: AI Agents & Decision Automation

SLOs Addressed: *SLO 1, SLO 3, SLO 5*

- What AI agents are: practical definition and current capabilities
- The agent loop: Perception → Reasoning → Action
- Where agents add value in business operations
- Limitations, risks, and appropriate use cases

Lab: Design a simple AI agent concept for monitoring or decision support. Students will create a design document for an AI agent including: purpose, inputs/sensors, decision logic, actions, human oversight requirements, and success metrics.

Module 10: AI, Ethics, Law & Governance (Applied)

SLOs Addressed: *SLO 4*

- Bias, fairness, and accountability as deployment risks
- Privacy considerations: data collection, consent, and protection
- Regulatory landscape: EU AI Act, US guidelines, industry standards
- Organizational responsibility and governance frameworks

Case Study Lab: Evaluate an AI ethics failure and propose mitigation steps. Students will analyze a documented AI ethics case (Amazon hiring, COMPAS recidivism, etc.) and develop a remediation plan including technical, process, and governance recommendations.

Module 11: AI in Government & Public Sector

SLOs Addressed: *SLO 2, SLO 3, SLO 4*

- Public sector workflows: service delivery, emergency response, resource allocation
- Smart city applications: traffic, utilities, public safety
- Transparency, explainability, and public accountability requirements
- Procurement and implementation challenges in government

Hands-On Activity: Assess an AI system used in public-sector decision-making. Students will evaluate a government AI application using criteria for transparency, fairness, effectiveness, and public trust, then present findings and recommendations.

Module 12: AI in Agriculture & Sustainability

SLOs Addressed: *SLO 2, SLO 3, SLO 5*

- Agricultural workflows: planting, monitoring, harvesting, distribution
- Precision agriculture: sensors, drones, satellite imagery, predictive analytics
- Environmental monitoring and sustainability applications
- Economic and environmental tradeoff analysis

Lab: Conceptual AI solution proposal for an agricultural workflow. Students will identify an agricultural challenge and develop a proposal including problem statement, AI approach, data requirements, implementation plan, and expected ROI.

Module 13: AI and Blockchain Integration

SLOs Addressed: *SLO 1, SLO 3, SLO 5*

- Introduction to blockchain technology: distributed ledgers, consensus mechanisms, smart contracts
- AI and blockchain synergies: data integrity, decentralized AI, secure model sharing
- Industry applications: supply chain traceability, healthcare records, financial services, identity verification
- Challenges and limitations: scalability, energy consumption, regulatory considerations

Lab: Evaluate an AI-blockchain use case. Students will analyze a real-world implementation combining AI and blockchain (e.g., supply chain provenance, decentralized AI marketplace, or healthcare data sharing), documenting the architecture, benefits, challenges, and lessons learned.

Module 14: Future Trends & Course Review

SLOs Addressed: *SLO 1, SLO 3, SLO 5*

- Emerging AI trends: copilots, agents, multimodal AI, edge computing
- Job evolution: how AI is changing roles across industries
- Recap of key concepts and frameworks from the course

- Continuous learning strategies for AI professionals

Activity: Career mapping: AI roles and required skills. Students will research AI job postings, identify common requirements, and create a personal development plan mapping their current skills to target roles.

Module 15: Student Presentations

SLOs Addressed: *SLO 5, SLO 6*

- Students presentations
- Industry panel feedback (when available)
- Peer review and constructive feedback sessions

Module 16: Finals Week

SLOs Addressed: *All SLOs (1-6)*

- Final Project due / Exam
- Course Portfolio due (GitHub)

Course Structure Notes

- Each module integrates process analysis + AI evaluation
- Emphasis on decision-making and business analysis, not coding
- Hands-on work mirrors real workplace tasks and deliverables
- Progressive build from understanding workflows to proposing AI solutions
- The curriculum is designed to prepare students for AI analyst and implementation roles

Artificial Intelligence

Artificial Intelligence is now a **core capability across nearly every industry**, including healthcare, manufacturing, energy, logistics, finance, retail, government, and education. AI is no longer limited to technology companies—it is widely used in **non-technical organizations** to automate processes, analyze data, monitor systems, and support decision-making.

The **Associate of Applied Science (AAS) in Artificial Intelligence & Robotics** is an **applied workforce degree** designed to prepare students for **entry-level and early-career AI roles** through hands-on experience with real-world tools and systems.

Some Entry-Level Job Roles

- AI Technician / AI Systems Specialist
- Applied AI or Junior AI Developer
- Data Analyst (AI-enabled)
- Machine Learning or Computer Vision Technician
- Automation & Intelligent Systems Specialist
- AI Operations (AIOps) Associate
- Healthcare AI Systems Support Specialist
- Manufacturing AI / Smart Factory Technician
- Logistics & Supply Chain AI Analyst

Employment Outlook

AI professionals are employed across **both technical and non-technical industries**, including hospitals, factories, utilities, transportation companies, insurance firms, and government agencies. AI-related roles are among the **fastest-growing job categories**, with strong demand for applied, entry-level talent.

Wages & Career Growth

- **Typical entry-level salaries:** \$65,000–\$90,000 annually
- **Mid-career applied AI roles:** \$90,000–\$140,000+
- Hourly technical roles often range from **\$35–\$75/hour**

Industry Shift

The workforce is shifting from traditional software-only roles toward **AI-enabled professionals** who can deploy, operate, and responsibly use AI systems in real-time

Student Clubs

- Computer Science Association (CSA) club dit.csa@hccs.edu
- Women in Technology (WIT) club <https://hccs.presence.io/organization/women-in-technology>

Process for Expressing Concerns about the Course

If you have concerns about any aspect of this course, please reach out to your instructor for assistance first. If your instructor is not able to assist you, then you may wish to contact the Department Chair.

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